Lecture 30 (*May 5*)

Reading: Section 12.2
Viewing: none

The true power of linear algebra in probability comes about when we run more than one experiment at once, and we want to compare the outcomes, like in the example of performing many coin tosses in the central limit theorem. The abstract concept that governs this is the following.

Definition 25. *A **random variable** is a quantity X that takes values in $\mathbb{R}$, which is either:*

- *discrete, so takes finitely many values $x_1, \ldots, x_n$ with probabilities $p_1, \ldots, p_n$, or*
- *continuous, associated to a probability distribution $p(x)$.*

In either case, the mean (or expected value) is denoted by:

$$E[X] = \mu \tag{275}$$

and is defined by either (267) or (271). Then the variance is:

$$E\Big[(X-\mu)^2\Big] = \Sigma \tag{276}$$

and is given by either (268) or (272).

Definition 26. *Given two random variables X and Y, their **covariance** is the quantity:*

$$\Sigma_{XY} = E\Big[(X - E[X])(Y - E[Y])\Big] \tag{277}$$

*Note that Σ_{XX} is just the **variance** of the random variable X itself.*

Let's start with the case of discrete probabilities, so X and Y both take a finite number of values with certain probabilities. The whole point of covariance is to measure the extent to which X and Y are correlated with each other. So it does not suffice to simply know the probabilities of the various values of X and Y separately, but instead you need to know the **joint probabilities**:

$$p_{ij} \text{ is the probability that } (X,Y) = (x_i, y_j) \tag{278}$$

where $x_1, x_2, \ldots, x_k$ are the possible values of X and $y_1, y_2, \ldots, y_k$ are the possible values of Y.

Example 8. *Let's assume that X and Y are coin flips, and their possible values are heads (H) and tails (T). If the two coin flips are independent, then the joint probabilities are:*

$$p_{HH} = p_{TT} = p_{HT} = p_{TH} = \frac{1}{4} \tag{279}$$

But if the random variables are two coin flips which are forced to always land on the same side (think something involving magnets), then the joint probabilities are:

$$p_{HH} = p_{TT} = \frac{1}{2} \quad \textit{and} \quad p_{HT} = p_{TH} = 0 \tag{280}$$

The means of two random variables X and Y are:

$$\mu = \sum_{i,j} p_{ij} x_i$$

$$\nu = \sum_{i,j} p_{ij} y_j$$

and therefore the covariance of the two random variables is:

$$\Sigma_{XY} = \sum_{i,j} p_{ij}(x_i - \mu)(y_j - \nu) \tag{281}$$

If X and Y are continuous random variables, their **joint probability distribution** is a function:

$$p(x, y) : \mathbb{R}^2 \to \mathbb{R}_{\geq 0} \tag{282}$$

which encodes probability in the sense that:

$$\text{Prob}(X \in [a, b] \text{ and } Y \in [c, d]) = \int_a^b \int_c^d p(x, y) dy dx \tag{283}$$

With this in mind, the covariance of the random variables X and Y is:

$$\Sigma_{XY} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x, y)(x - \mu)(y - \nu) dx dy \tag{284}$$

where μ and ν are the means of X and Y, defined by (271).

Remark. *The notion* (282) *is necessary because the random variables X and Y may be dependent on each other: mathematically, this means that where X lands on the real line might impact where Y lands. For example, if $X = Y$, then X and Y cannot be at different points of the real line, so in this case the function $p(x, y)$ is concentrated on the diagonal in $\mathbb{R}^2$.*

An important special case is when two variables X and Y are independent of each other. In the discrete case, this means that the probabilities (278) are given by $p_{ij} = a_i b_j$, where $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ are the probabilities of X and Y separately. In the continuous case, the fact that X and Y are independent means that the joint probability distribution (282) is given by:

$$p(x, y) = a(x)b(y)$$

where a and b are the probability distributions of X and Y, respectively. In this case, the position of X on the real line has no impact on the position of Y and vice versa. The covariance (283) is:

$$\Sigma_{XY} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} a(x)b(y)(x - \mu)(y - \nu) dx dy = \left(\int_{-\infty}^{\infty} a(x)(x - \mu) dx \right) \left(\int_{-\infty}^{\infty} b(y)(y - \nu) dy \right) = 0$$

(the last equality is a consequence of (270) and (271)). We conclude that the covariance of two <u>independent</u> random variables is 0. In general, we have the following principle:

- **the more positive the covariance, the more correlated the random variables**

- **the more negative the covariance, the more anti–correlated the random variables**

- **if the covariance is 0, the random variables are called $\boxed{\textbf{uncorrelated}}$**

For any random variables X and Y (discrete or continuous), the Cauchy-Schwartz inequality states:

$$|\Sigma_{XY}| \leq \sqrt{\Sigma_{XX}\Sigma_{YY}} \tag{285}$$

In other words, the covariance is bounded by the geometric mean of the variances. Therefore, if the variances of X and Y are fixed, the maximum covariance happens when $X = Y$ (the two random variables are as correlated as possible) and the minimum covariance happens when $X = -Y$ (the two random variables are as anti–correlated as possible).

Definition 27. *The covariance matrix of two random variables X and Y is:*

$$K = \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix} \tag{286}$$

Since $\Sigma_{XY} = \Sigma_{YX}$, the covariance matrix is symmetric. But it is actually <u>positive semidefinite</u>, because its trace is positive (as Σ_{XX} and Σ_{YY} are the variances of X and Y) and its determinant is ≥ 0 according to (285). To further analyze the covariance matrix, we have two sub-cases:

- *either* K is <u>positive definite</u>,

- *or* $\det K = 0$. When this happens, X and Y are called <u>perfectly correlated</u>.

Let's compute the covariance matrix of discrete random variables in terms of the probabilities (278):

$$K = \sum_{i,j=1}^{k} p_{ij} \begin{bmatrix} (x_i - \mu)^2 & (x_i - \mu)(y_j - \nu) \\ (x_i - \mu)(y_j - \nu) & (y_j - \nu)^2 \end{bmatrix} = \sum_{i,j=1}^{k} p_{ij} \underbrace{\begin{bmatrix} x_i - \mu \\ y_j - \nu \end{bmatrix} \begin{bmatrix} x_i - \mu & y_j - \nu \end{bmatrix}} \tag{287}$$

The underbraced term is a positive semidefinite symmetric matrix, since it is of the form $\boldsymbol{v}\boldsymbol{v}^T$. The p_{ij}'s are non-negative numbers, and it is easy to show that a non-negative linear combination of positive semidefinite matrices is positive semidefinite (for example, using the energy criterion (237)). This gives another proof of the fact that the covariance matrix is positive semidefinite.

Example 9. *Let's compute the covariance matrix in the case when X and Y are two coin tosses, as in Example 8. We will assign X and Y the value -1 if the coin falls on tails and 1 if the coin*

falls on heads. Clearly, both coin tosses have mean $\mu = \nu = 0$.

In the case of independent coin tosses (probabilities as in (279)*) we have:*

$$K = \frac{1}{4}\underbrace{\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}}_{outcome\ is\ HH} + \frac{1}{4}\underbrace{\begin{bmatrix}-1\\-1\end{bmatrix}\begin{bmatrix}-1 & -1\end{bmatrix}}_{outcome\ is\ TT} + \frac{1}{4}\underbrace{\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix}}_{outcome\ is\ HT} + \frac{1}{4}\underbrace{\begin{bmatrix}-1\\1\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix}}_{outcome\ is\ TH} = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

The covariance matrix is diagonal because the random variables are independent, hence uncorrelated.

In the case of identical coin tosses (probabilities as in (280)*) we have:*

$$K = \frac{1}{2}\underbrace{\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}}_{outcome\ is\ HH} + \frac{1}{2}\underbrace{\begin{bmatrix}-1\\-1\end{bmatrix}\begin{bmatrix}-1 & -1\end{bmatrix}}_{outcome\ is\ TT} + 0\underbrace{\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix}}_{outcome\ is\ HT} + 0\underbrace{\begin{bmatrix}-1\\1\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix}}_{outcome\ is\ TH} = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$$

The covariance matrix has determinant 0 because identical random variables are perfectly correlated.